

Rolling Root Mean Square Based Multimodal Anomaly Detection for Real Time Monitoring of Smart Grid

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Abstract—Reliable real-time monitoring is valuable for maintaining the operational integrity of modern electrical smart grids. Deployment of heterogeneous sensing technologies in substations has enabled high-resolution, multichannel waveform monitoring, but also introduces challenges for anomaly detection due to noise, baseline drift, and modality-dependent signal characteristics. In this work, we present a computationally efficient unsupervised method for multimodal event detection based on Rolling Root Mean Square based Event Detection (RRMSED). The method is developed using in-house, field deployed sensors collecting data at a utility substation. The sensing system comprises voltage and current sensors, triaxial accelerometers, and magnetometers, collectively capturing electrical, vibrational, and magnetic waveform measurements at high temporal resolution. RRMSED operates by extracting rolling RMS energy features and their first-order temporal differences from consecutive waveform segments for each channel and then applying channel-specific statistical thresholds learned from historical data. A persistence-based exceedance logic is employed to robustly identify transient events while suppressing impulsive noise, and to provide precise temporal localization with high resolution. The framework is designed for continuous server-side operation and can be deployed in real time without requiring complex models. Experiments on simulated waveform data with known ground truth demonstrate low false positive (FP) and false negative (FN) rates. Application to real substation data shows RRMSED to identify events that are not captured by conventional monitoring indicators including fast transient detection algorithm currently deployed in the system. These results indicate that rolling RMS based features provide an effective and practical basis for real-time multimodal event detection in smart-grid substations.

Index Terms—Multimodal sensing data, event detection, grid monitoring, rolling root mean square, false positive, false negative.

I. INTRODUCTION

Deployment of various sensing technologies in modern electrical substations can enable continuous, high-resolution monitoring and diagnostics of grid infrastructure. Multichannel waveform data acquired from heterogeneous sensors provide rich information regarding mechanical, magnetic, and electrical phenomena occurring within the system. Accurate and timely detection of transient events in such data is critical for ensuring operational reliability, preventing equipment

failure, and supporting informed decision-making in safety-critical environments. [1], [2] As substations evolve toward data-intensive smart grid architectures, the need for robust, interpretable, and computationally efficient event detection algorithms has become increasingly pronounced [3]–[6].

Traditional anomaly detection approaches face several limitations in multivariate sensing environments. Supervised and semi-supervised learning methods often suffer from severe class imbalance and the scarcity of labeled anomalous events. The manual annotation of high-frequency waveform data is labor-intensive and frequently impractical in operational settings [2]. Unsupervised deep learning models, including recurrent, convolutional, and generative architectures, have demonstrated strong performance in modeling complex temporal dependencies [4]–[6]. However, their black-box nature limits interpretability, and their computational overhead restricts real-time deployment in resource-constrained environments. Furthermore, their sensitivity to distribution shifts and unseen operating conditions can reduce reliability in dynamic grid systems. In practical grid environments, transient disturbances may be subtle, short-lived, and modality-dependent, requiring detection mechanisms that are both sensitive to localized energy changes and robust to baseline drift.

To address these challenges, this work focuses on real-time event detection in multimodal substation waveform data using the RRMSED framework, which is designed to operate continuously on a server in real time without reliance on complex models or adaptive parameter tuning. Conventional transient detection algorithms are typically tuned to capture high-amplitude or sharply impulsive disturbances. However, subtle, short-duration waveform distortions may remain undetected under such configurations. A primary objective of this study is therefore to investigate whether rolling RMS-based statistical analysis can identify transient behaviors that are not captured by conventional fast transient detection methods.

The primary contributions of this work are summarized as follows:

- **Multimodal Data Acquisition:** We deploy a heterogeneous sensing suite at a commercial utility substation

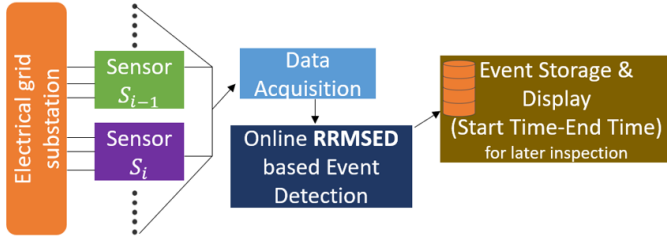


Fig. 1: Overview of the proposed RRMSED framework. S_i denotes the multimodal sensing suite deployed at the electrical substation. The pipeline includes data acquisition, feature extraction, threshold calibration, and online event detection.

(EPB Big Ridge near Chattanooga, TN), consisting of voltage and current sensors, two triaxial accelerometers, and two triaxial magnetometers, collectively capturing electrical, vibrational, and magnetic waveform measurements at high temporal resolution.

- **RRMSED Framework:** We develop a computationally efficient and interpretable event detection framework based on rolling RMS energy analysis. The method operates in two stages: (i) a calibration stage, where channel-specific statistical thresholds are estimated from historical data, and (ii) an online inference stage, where incoming waveform segments are processed for event detection and precise temporal localization. The implementation of RRMSED is publicly available [7].
- **Controlled Performance Evaluation:** To enable quantitative assessment, we generate synthetic waveform datasets that mimic the statistical and spectral properties of real substation signals while allowing controlled injection of events with known onset and termination times. On these datasets, RRMSED demonstrates low false positive (FP) and false negative (FN) rates. Application to experimental substation data further illustrates the ability of the framework to detect transient events beyond those captured by conventional fast transient [3], [8] indicators.

Figure 1 presents a schematic overview of the proposed framework. The remainder of the paper is organized as follows. Section II describes the multimodal sensing setup and data acquisition process at the EPB Big Ridge substation. Section III details the RRMSED methodology. Section IV introduces the synthetic data generation procedure for controlled evaluation. Section V presents experimental results on both simulated and real substation data. Section VI concludes the paper and outlines directions for future research.

II. EXPERIMENTAL DATA COLLECTION

Figure 2 illustrates the multimodal sensing configuration deployed at the EPB Big Ridge substation transformer. The experimental setup integrates magnetic, vibrational, and electrical measurements to enable comprehensive monitoring of transformer behavior under operational conditions. Magnetic field

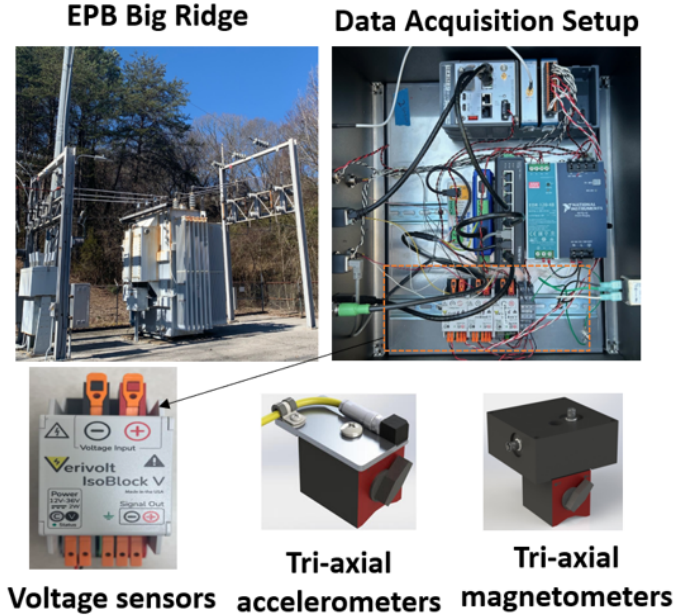


Fig. 2: Multimodal sensing configuration deployed at the EPB Big Ridge substation transformer, including triaxial accelerometers, triaxial magnetometers, voltage monitoring channels, and associated signal conditioning hardware.

measurements are acquired using two triaxial magnetometers (T2-2 and T2-4), each capturing variations along three orthogonal directions (X, Y, and Z). These sensors provide insight into electromagnetic fluctuations associated with load variations and transient disturbances. Electrical measurements are obtained from a single phase voltage monitoring channel of the transformer. A Verivolt IsoBlock signal conditioning module is employed to provide electrical isolation and voltage scaling, converting the potential-transformer (PT) voltage (approximately 120 VAC) to a reduced level suitable for data-acquisition equipment. This module serves as an interface between the transformer PT and the data acquisition system and does not constitute a sensing modality itself. Mechanical vibrations are recorded using two triaxial accelerometers (T2-1 and T2-3), measuring acceleration along the X, Y, and Z axes. These measurements capture structural and electromechanical responses of the transformer during steady-state and transient operating conditions. All sensor signals are recorded using a NI CompactRIO (cRIO) data acquisition system at a sampling rate of 20 kHz for the vibrational and voltage channels, and 10 kHz for the magnetic field channels.

III. METHODOLOGY

Figure 3 illustrates the working principle of the RRMSED algorithm, which operates in two phases: (i) a calibration stage and (ii) an online inference stage. The framework extracts rolling RMS-based statistical features from consecutive waveform segments and applies channel-specific statistical

thresholds to identify transient energy variations. Importantly, the feature extraction pipeline is identical for historical and streaming data; only the aggregation of historical feature statistics is performed once to construct a global threshold lookup table.

For a given sensing channel, the incoming data stream is partitioned into consecutive waveform segments of approximately 2 seconds. Let $x_n[m]$, where $m = 0, 1, \dots, N-1$, denote the discrete-time samples of the n -th segment consisting of N samples, with the sampling frequency inferred as $f_s = N/2$.

To mitigate boundary effects in rolling statistics and preserve temporal continuity, adjacent segments are conditionally augmented using a patch-based stitching strategy. Let $\tilde{x}_n[m]$ denote the (optionally) patched waveform corresponding to segment n . When two successive segments are separated by approximately 2000 ms (within a tolerance of 20 ms) and have equal length, a short tail from the previous segment is appended to the current segment. The resulting patch offset is retained for accurate index-to-time mapping during event localization. When these conditions are not satisfied, no patching is applied and $\tilde{x}_n[m] = x_n[m]$, with the patch offset set to zero.

Baseline drift and slow-varying offsets commonly observed in field-deployed sensors are removed using rolling DC offset subtraction. Specifically, a rolling mean $\mu_n[m]$ computed over a window of length $N_w = \lfloor N/\text{WINDOW_FRACTION} \rfloor$ (with $\text{WINDOW_FRACTION} = 20$) is subtracted from $\tilde{x}_n[m]$ to obtain the zero-mean waveform $\hat{x}_n[m] = \tilde{x}_n[m] - \mu_n[m]$. For 2-second segments and $\text{WINDOW_FRACTION} = 20$, the RRMS window corresponds to approximately 100 ms.

Local signal energy is quantified using a rolling Root Mean Square (RMS) statistic defined as $R_n[t] = \sqrt{\frac{1}{N_w} \sum_{i=t}^{t+N_w-1} \hat{x}_n^2[i]}$. Abrupt energy variations are emphasized through the rolling RMS-difference $\Delta R_n[t] = R_n[t + N_w] - R_n[t]$. To normalize across channels with different amplitude scales, a standardized RMS-difference feature is defined as $S_n[t] = \left| \frac{\Delta R_n[t]}{R_n[t]} \right|$. Depending on the detection mode, RRMSED operates on either $|\Delta R_n[t]|$ or $S_n[t]$. In this manuscript, the standardized RMS-difference feature is used for event detection.

From each waveform segment, a scalar summary statistic is extracted as the maximum standardized RMS-difference within the segment, denoted $\phi_n = \max_t S_n[t]$. During the calibration stage, these segment-level statistics are accumulated to estimate the empirical mean μ and standard deviation σ for each sensing channel. An upper control limit for the standardized RMS-difference is then defined as $\tau = \mu + k\sigma$, where $k = 3$ is used as the default value for this tuning parameter and larger values provide more conservative detection. The resulting thresholds are stored as channel-specific lookup tables and reused during online operation.

The suitability of a fixed global threshold was assessed

using the Augmented Dickey–Fuller test [9] on the extracted feature sequences. Across all channels and selected features, the null hypothesis of non-stationarity was rejected at the 5% significance level, indicating statistical stationarity of the feature distributions. Under this condition, a fixed statistical threshold deemed appropriate and avoids the additional complexity associated with adaptive thresholding methods.

During online inference, incoming waveform segments are processed using the same feature extraction pipeline. Let $F_n[t]$ denote the selected feature signal for segment n . Event onset is declared when $F_n[t]$ exceeds the channel-specific threshold τ for three consecutive samples, while event termination is declared when the feature remains below threshold for three consecutive samples. Although the decision is made upon observing the third consecutive exceedance (or non-exceedance), the reported onset and termination indices correspond to the first sample in the respective three-sample sequence.

Detected events are initially identified in feature-index space and subsequently mapped to waveform indices to obtain physical timestamps. Because rolling statistics introduce a deterministic index shift, feature indices $(i_{\text{start}}, i_{\text{end}})$ are converted to waveform indices according to $j_{\text{start}} = i_{\text{start}} + N_w - \text{patch_offset}$ and $j_{\text{end}} = i_{\text{end}} + N_w - \text{patch_offset}$. When patching is not applied, the patch offset is zero and the same mapping remains valid. Waveform indices are clipped to the bounds of the current segment, and absolute event timestamps are computed as $t_{\text{start}} = t_n + \frac{j_{\text{start}}}{f_s}$ and $t_{\text{end}} = t_n + \frac{j_{\text{end}}}{f_s}$, where t_n denotes the recorded start time of the segment.

Multiple events may occur within a single segment or span across adjacent segments. Events separated by less than 50 ms are merged first within each segment and subsequently across segment boundaries using absolute timestamps. For each merged event, the peak feature value across all constituent sub-events is retained.

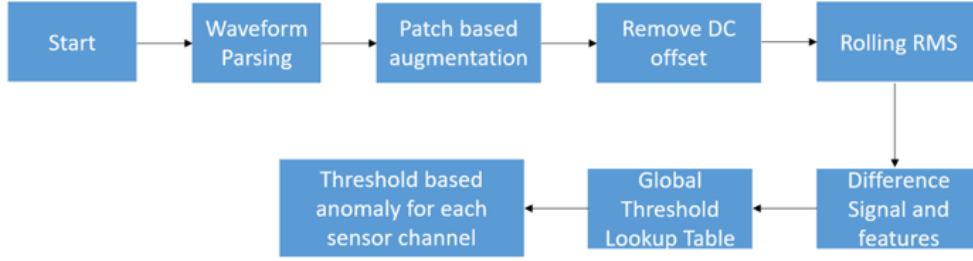
Each finalized event is stored as a structured record containing the channel group, sensor identifier, detection method, peak feature value, applied threshold, and event start and end timestamps with microsecond precision. An example of such a stored event record is presented in Table I.

IV. SIMULATED DATA

A key limitation of the field-deployed sensing system is the absence of sample-level ground truth labels for transient events in the server data. Consequently, FP and FN rates cannot be directly quantified on the real dataset. To enable controlled performance evaluation of the proposed Rolling Root Mean Square–Based Event Detection (RRMSED) framework, we generated synthetic waveform datasets that mimic the statistical and spectral characteristics of the real voltage signals while allowing explicit injection of events with known onset and termination times.

Synthetic normal voltage waveforms were generated using a template-based procedure derived from short representa-

Phase I: Calibration time feature extraction and global threshold construction



Phase II: Online Inference

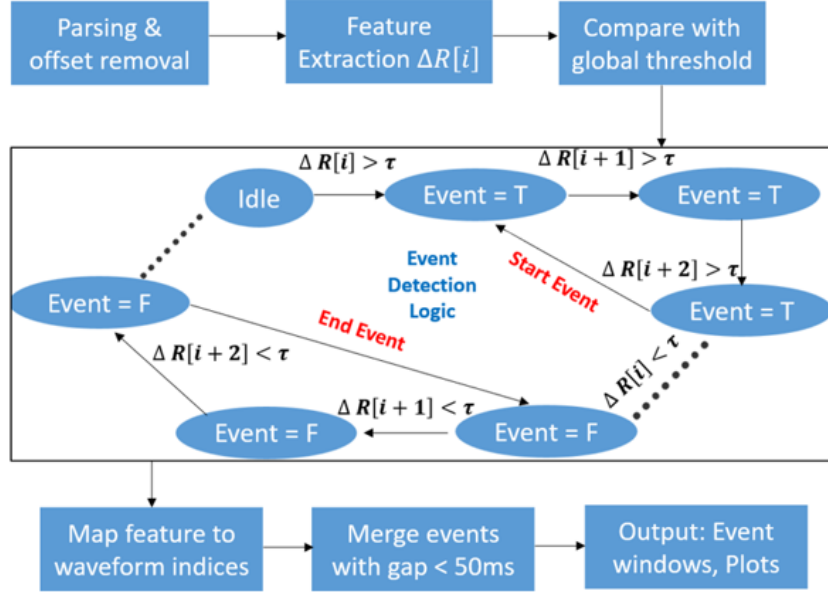


Fig. 3: Working principle of the RRMSED algorithm in two phases: (i) calibration stage for feature aggregation and threshold construction, and (ii) online inference stage for event detection and localization.

TABLE I: Example of an event record generated by RRMSED

channel_group	channel	method	value	threshold	start_time	end_time
EPB-BIG	EPB.BIG.PT1.Voltage.110	RRMS	0.0106	0.0068	1725534318	1725534319

tive snippets of real substation recordings. From these snippets, dominant frequency components, carrier amplitude, noise level, and slow baseline drift characteristics were estimated. A continuous synthetic signal was then constructed by combining a sinusoidal carrier with randomized phase, low-frequency drift, and additive noise matching the statistics of the real data. The continuous waveform was segmented into consecutive 2-second windows identical to the server representation and stored in the same processed CSV format.

Transient events were injected into the continuous waveform prior to segmentation so that their exact start and end times were known. Events were parameterized by (i) an arrival model, (ii) event type, (iii) duration, and (iv) strength. Supported event types included amplitude steps, bursts, ringdowns, spike trains, and variance changes. For each injected event,

ground truth metadata including file identifier and absolute start and end times were recorded to enable later comparison with detector outputs.

To emulate the operational calibration procedure, 60 synthetic files without injected events were used to compute channel-specific thresholds based on rolling RMS-derived features. These thresholds were then applied to a separate test dataset consisting of 50 synthetic files containing 49 injected events with known locations. RRMSED was executed using the calibration-derived threshold, and detected event intervals were compared against injected intervals to assess FN and FP. Quantitative evaluation results are presented in Section V.

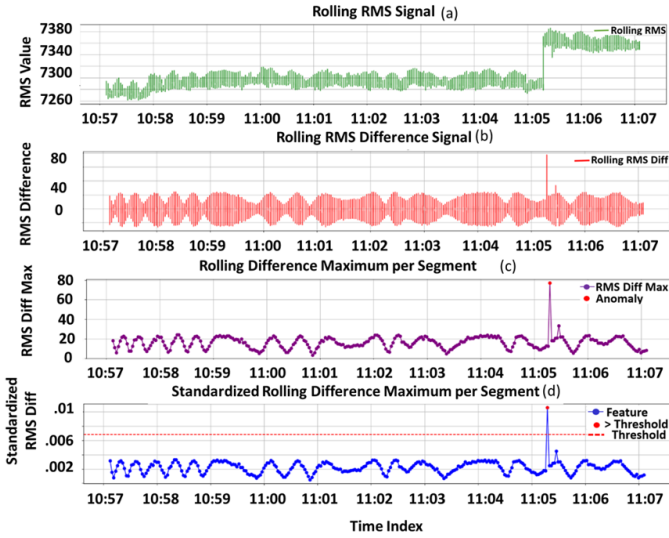


Fig. 4: Feature extraction steps of the RRMSD algorithm applied to experimental voltage data: (a) rolling RMS signal, (b) rolling RMS-difference, (c) segment-wise maximum RMS-difference, and (d) segment-wise maximum standardized RMS-difference.

V. RESULTS & DISCUSSION

A. Results on Experimental Data

To demonstrate the behavior of the proposed RRMSD framework on field measurements, we analyze a representative voltage channel (EPB.BIG.PT1.Voltage.110) extracted from a large processed server file (approximately 3 GB, corresponding to 10 minutes of continuous recording). Figure 4 illustrates the intermediate feature extraction steps of the RRMSD algorithm. Figure 4(a) shows the rolling RMS signal, while Figure 4(b) presents the corresponding rolling RMS-difference. Both quantities exhibit localized spikes corresponding to abrupt changes in signal energy. Figures 4(c) and 4(d) display the segment-wise statistical features computed over consecutive 2-second fragments, namely the maximum RMS-difference and the maximum standardized RMS-difference, respectively. Segments whose feature values exceed the calibrated global threshold (evaluated with $k = 3$) are marked, indicating candidate anomalous intervals.

Using the persistence-based logic described in Section III, event start and end times are subsequently inferred. Figure 5 presents a localized visualization of one detected event. The inferred start and termination times are indicated by vertical markers. For qualitative interpretation, a zoomed waveform segment centered at $t_c = (t_{\text{start}} + t_{\text{end}})/2$ is extracted within a temporal window $[t_c - \Delta, t_c + \Delta]$, where Δ is a user-selected half-window duration (e.g., 0.5 s). The zoomed view clearly reveals a distortion in the original waveform corresponding to the detected transient.

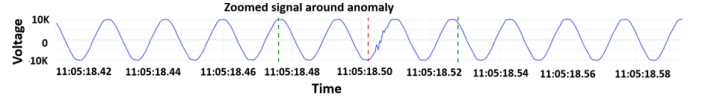


Fig. 5: Zoomed visualization of a detected event in the voltage waveform. Green vertical markers indicate the inferred start and end times of the event, while the red marker denotes the midpoint between the start and end times.

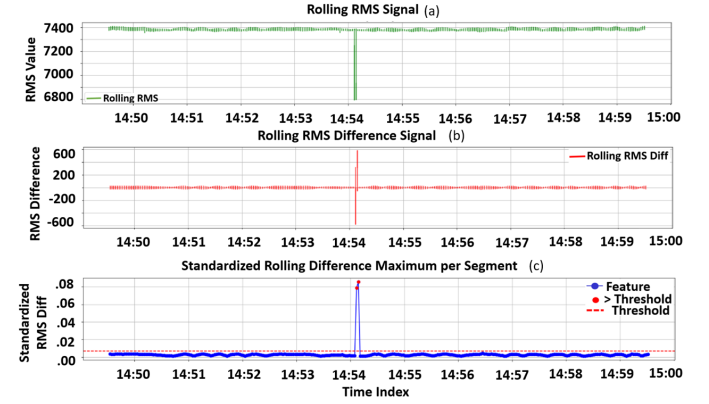


Fig. 6: Feature extraction results illustrating two transient segments detected by RRMSD but not flagged by the conventional fast transient algorithm.

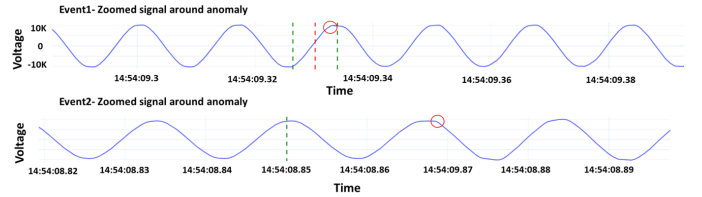


Fig. 7: Zoomed voltage waveforms corresponding to the additional events detected by RRMSD. Subtle waveform distortions are highlighted with red circles, and vertical markers indicate the inferred start and end times of the respective events.

It is noteworthy that the event shown in Figure 5 was also identified by the existing fast transient detection algorithm currently deployed in the monitoring system. Figure 6 illustrates two events detected by RRMSD but not flagged by the fast transient algorithm within the same 10-minute recording. The standardized RMS-difference feature in Figure 6(c) highlights two distinct segments exceeding the global threshold. Corresponding zoomed waveform views are shown in Figure 7, where subtle distortions or kinks in the voltage waveform are visible. Although these transient behaviors are less pronounced than classical fast transients, their consistent detection by RRMSD suggests sensitivity to localized energy variations that may otherwise remain undetected.

Further investigation, including expert domain assessment, is required to determine whether these additional detected

transients correspond to benign operational variations or events of potential interest. Nevertheless, these observations indicate that RRMSD provides complementary detection capability relative to conventional fast transient methods while maintaining interpretability and computational efficiency suitable for real-time deployment.

B. Results on Simulated Data

Using 60 event-free synthetic files as calibration data, we computed the channel-specific global threshold for the standardized RMS-difference feature using $k = 6$ and applied it to a separate test set of 50 synthetic files containing 49 injected events. Detected events were considered *matched* to ground truth if both the estimated start and end times fell within ± 1 s of the corresponding ground truth times. As summarized in Table IV (see next page), RRMSD successfully matched 47 out of 49 injected events. The remaining two injected events were missed FN, consisting of one `ringdown` and one `variance_change` event (Table II). In addition, one spurious detection was produced with no corresponding injected event FP, as reported in Table III.

We repeated the same analysis using a less conservative threshold ($k = 3$). The number of matched events remained 47 with 2 FN; however, the number of FP increased to 7, illustrating the expected trade-off controlled by the threshold multiplier k . Here, a FP corresponds to a detected event in the absence of a true transient, while a FN corresponds to a missed detection of an actual event. This trade-off is illustrated schematically in Figure 8.

TABLE II: FN by the algorithm with $k = 6$.

Start Det.	End Det.	Parent csv	Event Type	Value
11:28:10.783	11:28:11.401	synthetic_5	variance_change	0.202
11:44:08.847	11:44:09.428	synthetic_29	ringdown	0.314

TABLE III: FP by algorithm with $k = 6$.

Parent csv	Value	Threshold	Start Det.	End Det.
synthetic_49	0.223	0.0424	11:57:15.833	11:57:16.006

		Truth	
		Anomaly	Non-Anomaly
Detected by Algorithm	Anomaly	Correct Detection	False Positive
	Non-Anomaly	False Negative	Correct Detection

Fig. 8: Schematic representation of detection outcomes in a threshold-based anomaly detection framework.

VI. CONCLUSION & FUTURE WORK

In this work, we proposed the RRMSD framework for real-time multimodal event detection in substation waveform data. The method leverages rolling RMS energy features and a persistence-based thresholding mechanism to identify localized transient disturbances in high-frequency multichannel signals. Its computational simplicity and interpretability make it suitable for safety-critical monitoring environments.

Controlled evaluation on synthetic waveform datasets with known ground truth demonstrated that RRMSD achieves low FP and FN rates when using thresholds calibrated from event-free data. These results validate the effectiveness of the rolling RMS based statistical features for transient event detection under controlled conditions.

Application to experimental substation data further illustrates the practical capability of RRMSD. In addition to detecting events also captured by the existing fast transient detection algorithm, RRMSD identified additional transient behaviors that were not flagged by conventional methods. While these additional detections indicate increased sensitivity to localized energy variations, the absence of sample level ground truth labels in the real server data prevents direct quantitative assessment of FP and FN rates on experimental recordings. Establishing reliable ground truth for real substation events, including structured comparison with supervisory control and data acquisition (SCADA)-based indicators, remains an important direction for future validation.

Future work will focus on systematic expert evaluation of the additional detected transients to determine whether they correspond to benign operational variations or anomalies of practical significance. Although the present study reports detailed results for the voltage channel, the RRMSD framework operates independently on each sensing modality; therefore, future work will extend the analysis to acceleration and magnetic field channels and investigate fusion of information across sensing modalities. We will also examine adaptive sensing or channel-prioritization strategies that allocate enhanced scrutiny to the most informative sensors during emerging disturbances. The results indicate that rolling RMS based statistical analysis provides an effective and deployment ready foundation for multimodal event detection in smart grid substations.

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TABLE IV: Events matched by RRMSED on synthetical data ($k = 6$ is chosen for threshold). Here `Start GT` and `End gt` refer to the ground truth event times, while `Start Det.` and `End Det.` are the times detected by the algorithm. The columns `Start Err (s)` and `End Err (s)` indicate that the matching errors are within 1 second.

#	Event Type	Value	Threshold	Start GT	End GT	Start Det.	End Det.	Start Err (s)	End Err (s)
1	amp_step	0.2321	0.0424	11:25:35.464	11:25:35.544	11:25:35.380	11:25:35.629	0.0834	0.0842
2	amp_step	0.2775	0.0424	11:32:05.128	11:32:05.231	11:32:05.043	11:32:05.318	0.0846	0.0865
3	spike_train	9.9379	0.0424	11:32:11.214	11:32:11.795	11:32:11.114	11:32:11.894	0.0999	0.0990
4	amp_step	0.3007	0.0424	11:32:44.551	11:32:44.711	11:32:44.461	11:32:44.800	0.0892	0.0888
5	amp_step	0.2372	0.0424	11:33:23.676	11:33:23.776	11:33:23.593	11:33:23.855	0.0830	0.0795
6	amp_step	0.6428	0.0424	11:33:48.056	11:33:48.282	11:33:48.000	11:33:48.375	0.0563	0.0933
7	amp_step	0.6424	0.0424	11:34:12.240	11:34:12.511	11:34:12.142	11:34:12.608	0.0981	0.0970
8	amp_step	0.1228	0.0424	11:34:26.177	11:34:26.417	11:34:26.125	11:34:26.495	0.0521	0.0772
9	amp_step	0.1293	0.0424	11:34:58.091	11:34:58.195	11:34:58.026	11:34:58.250	0.0656	0.0549
10	spike_train	26.7144	0.0424	11:35:53.544	11:35:53.840	11:35:53.418	11:35:53.900	0.1257	0.0594
11	spike_train	11.4132	0.0424	11:35:54.453	11:35:54.878	11:35:54.354	11:35:54.971	0.0986	0.0927
12	amp_step	0.1757	0.0424	11:36:25.394	11:36:25.586	11:36:25.322	11:36:25.665	0.0722	0.0797
13	amp_step	0.4005	0.0424	11:36:51.426	11:36:51.536	11:36:51.332	11:36:51.624	0.0946	0.0877
14	amp_step	0.3966	0.0424	11:38:01.418	11:38:01.560	11:38:01.330	11:38:01.655	0.0886	0.0953
15	amp_step	0.4883	0.0424	11:38:10.520	11:38:10.712	11:38:10.428	11:38:10.804	0.0923	0.0922
16	amp_step	0.1412	0.0424	11:38:52.043	11:38:52.173	11:38:52.000	11:38:52.246	0.0437	0.0732
17	amp_step	0.4859	0.0424	11:39:37.068	11:39:37.360	11:39:36.974	11:39:37.455	0.0938	0.0950
18	ringdown	0.4060	0.0424	11:41:17.969	11:41:18.893	11:41:17.872	11:41:18.266	0.0977	0.6275
19	amp_step	0.6014	0.0424	11:41:27.472	11:41:27.606	11:41:27.379	11:41:27.699	0.0934	0.0934
20	amp_step	0.3185	0.0424	11:42:12.250	11:42:12.319	11:42:12.161	11:42:12.411	0.0891	0.0922
21	amp_step	0.5452	0.0424	11:42:18.885	11:42:19.061	11:42:18.790	11:42:19.153	0.0959	0.0928
22	amp_step	0.4159	0.0424	11:44:07.095	11:44:07.281	11:44:07.000	11:44:07.372	0.0950	0.0917
23	amp_step	0.1229	0.0424	11:26:40.997	11:26:41.124	11:26:40.938	11:26:41.197	0.0586	0.0727
24	amp_step	0.4188	0.0424	11:45:38.938	11:45:39.093	11:45:38.847	11:45:39.184	0.0909	0.0909
25	amp_step	0.5607	0.0424	11:45:57.749	11:45:58.046	11:45:57.653	11:45:58.142	0.0958	0.0954
26	amp_step	0.1750	0.0424	11:46:16.384	11:46:16.614	11:46:16.309	11:46:16.686	0.0748	0.0724
27	amp_step	0.3236	0.0424	11:46:48.980	11:46:49.207	11:46:48.895	11:46:49.294	0.0856	0.0874
28	amp_step	0.1893	0.0424	11:48:50.674	11:48:50.758	11:48:50.593	11:48:50.842	0.0808	0.0840
29	amp_step	0.3938	0.0424	11:49:55.806	11:49:55.906	11:49:55.719	11:49:55.900	0.0871	0.0068
30	amp_step	0.3348	0.0424	11:27:02.781	11:27:03.071	11:27:02.691	11:27:03.153	0.0897	0.0821
31	amp_step	0.6099	0.0424	11:51:21.884	11:51:21.973	11:51:21.791	11:51:22.068	0.0935	0.0956
32	spike_train	18.9878	0.0424	11:51:25.651	11:51:25.915	11:51:25.545	11:51:26.009	0.1056	0.0941
33	amp_step	0.6342	0.0424	11:54:48.779	11:54:48.893	11:54:48.685	11:54:48.986	0.0945	0.0935
34	amp_step	0.5658	0.0424	11:55:21.143	11:55:21.364	11:55:21.050	11:55:21.457	0.0925	0.0938
35	spike_train	8.7375	0.0424	11:55:33.156	11:55:33.319	11:55:33.066	11:55:33.407	0.0897	0.0873
36	amp_step	0.1981	0.0424	11:56:41.410	11:56:41.678	11:56:41.332	11:56:41.758	0.0786	0.0802
37	variance_change	0.1785	0.0424	11:57:04.514	11:57:04.694	11:57:04.437	11:57:04.764	0.0773	0.0702
38	variance_change	0.3884	0.0424	11:57:15.298	11:57:15.914	11:57:15.207	11:57:15.378	0.0912	0.5361
39	amp_step	0.5493	0.0424	11:57:42.946	11:57:43.047	11:57:42.854	11:57:43.144	0.0923	0.0971
40	amp_step	0.5062	0.0424	11:28:31.299	11:28:31.371	11:28:31.205	11:28:31.464	0.0936	0.0933
41	variance_change	0.1957	0.0424	11:28:32.684	11:28:32.823	11:28:32.598	11:28:32.890	0.0857	0.0675
42	ringdown	0.2124	0.0424	11:29:03.126	11:29:04.126	11:29:03.035	11:29:03.328	0.0915	0.7983
43	burst	0.1683	0.0424	11:29:10.879	11:29:11.304	11:29:10.922	11:29:11.260	0.0433	0.0439
44	amp_step	0.5620	0.0424	11:29:59.530	11:29:59.820	11:29:59.437	11:29:59.900	0.0931	0.0792
45	amp_step	0.3250	0.0424	11:30:13.937	11:30:14.235	11:30:13.841	11:30:14.326	0.0952	0.0914
46	amp_step	0.5465	0.0424	11:30:49.071	11:30:49.271	11:30:48.978	11:30:49.363	0.0934	0.0917
47	amp_step	0.4487	0.0424	11:30:58.548	11:30:58.723	11:30:58.454	11:30:58.813	0.0935	0.0898

provide public access to these results of federally sponsored research in accordance with the DOE Public Access Plan (<http://energy.gov/downloads/doepublic-access-plan>

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